

Formation of Binary Neutron Star

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Pulsar Coffee | Aug. 2023



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(PAST WORK) SHADOW OF COMPACT OBJECTS



Center of the galaxy M87
Captured by the EHT(2019).

What do we know about DNS systems so far?

- There are 17 known galactic BNS with total mass ranging from 2.50–2.89 M
- 12 has mass measured for both components- implying chirp mass of 1.12–1.24 M
- Two GW merger detections
 - GW170817
 - GW190425

Table 1
Observed Ranges of Key Properties of DNS Systems

Properties of Recycled (Old) NSs:	
Spin period, P	23–185 ms
Period derivative, $\dot{P}$	$(0.027\text{--}18) \times 10^{-18} \text{ s s}^{-1}$
Surface dipole B -field, B	$(0.29\text{--}18) \times 10^9 \text{ G}$
Mass, $M_{\text{NS},1}$	$1.32\text{--}1.56 M_{\odot}^{\text{a}}$
Properties of Young NSs:	
Spin period, P	144–2773 ms
Period derivative, $\dot{P}$	$(0.89\text{--}20) \times 10^{-15} \text{ s s}^{-1}$
Surface dipole B -field, B	$(2.7\text{--}5.3) \times 10^{11} \text{ G}$
Mass, $M_{\text{NS},2}$	$1.17\text{--}1.39 M_{\odot}$
Orbital Properties:	
Orbital period, P_{orb}	0.10–45 days
Eccentricity, e	0.085–0.83
Merger time, τ_{gw}	86 Myr $\rightarrow \infty$
Systemic velocity, v_{sys}	25–240 km s $^{-1}$

Note. Data taken from the ATNF Pulsar Catalogue (Manchester et al. 2005)—see Table 2 for further details. Only DNS systems in the Galactic disk are listed. The systemic recoil velocity, $v_{\text{sys}} = v^{\text{LSR}}$, is quoted with respect to the local standard of rest (Section 2.2).

^a $1.32 M_{\odot}$ Mark an upper limit to the lowest mass of the first-born NS.

Table 1
Source Properties for GW190425

	Low-spin Prior ($\chi < 0.05$)	High-spin Prior ($\chi < 0.89$)
Primary mass m_1	$1.60\text{--}1.87 M_\odot$	$1.61\text{--}2.52 M_\odot$
Secondary mass m_2	$1.46\text{--}1.69 M_\odot$	$1.12\text{--}1.68 M_\odot$
Chirp mass $\mathcal{M}$	$1.44^{+0.02}_{-0.02} M_\odot$	$1.44^{+0.02}_{-0.02} M_\odot$
Detector-frame chirp mass	$1.4868^{+0.0003}_{-0.0003} M_\odot$	$1.4873^{+0.0008}_{-0.0006} M_\odot$
Mass ratio m_2/m_1	$0.8 - 1.0$	$0.4 - 1.0$
Total mass m_{tot}	$3.3^{+0.1}_{-0.1} M_\odot$	$3.4^{+0.3}_{-0.1} M_\odot$
Effective inspiral spin parameter χ_{eff}	$0.012^{+0.01}_{-0.01}$	$0.058^{+0.11}_{-0.05}$
Luminosity distance D_L	$159^{+69}_{-72} \text{ Mpc}$	$159^{+69}_{-71} \text{ Mpc}$
Combined dimensionless tidal deformability $\tilde{\Lambda}$	≤ 600	≤ 1100

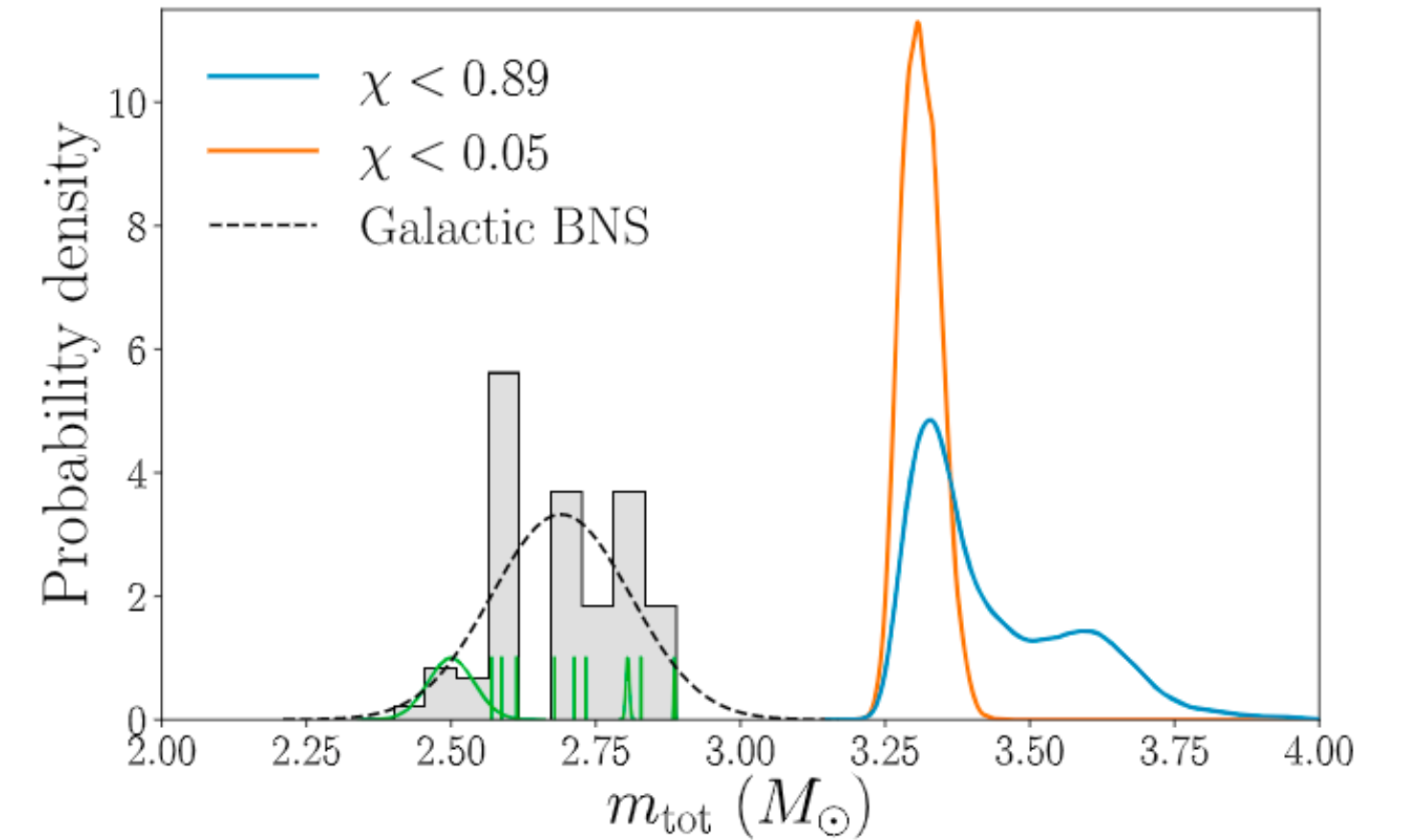


Figure 5. Total system masses for GW190425 under different spin priors, and those for the 10 Galactic BNSs from Farrow et al. (2019) that are expected to merge within a Hubble time. The distribution of the total masses of the latter is shown and fit using a normal distribution shown by the dashed black curve. The green curves are for individual Galactic BNS total mass distributions rescaled to the same ordinate axis height of 1.

B.P. Abbot et. al (2020)

Clear indication that GW190425 WAS FORMED DIFFERENTLY!

Known Formation channels for DNS systems

ISOLATED

Standard formation channel for Galactic field BNS: Two neutron stars are formed in a sequence of supernova explosions that occur in an isolated binary

DYNAMIC

Systems formed in dense stellar cluster

Standard DNS Formation Pathway

Tauris et al.

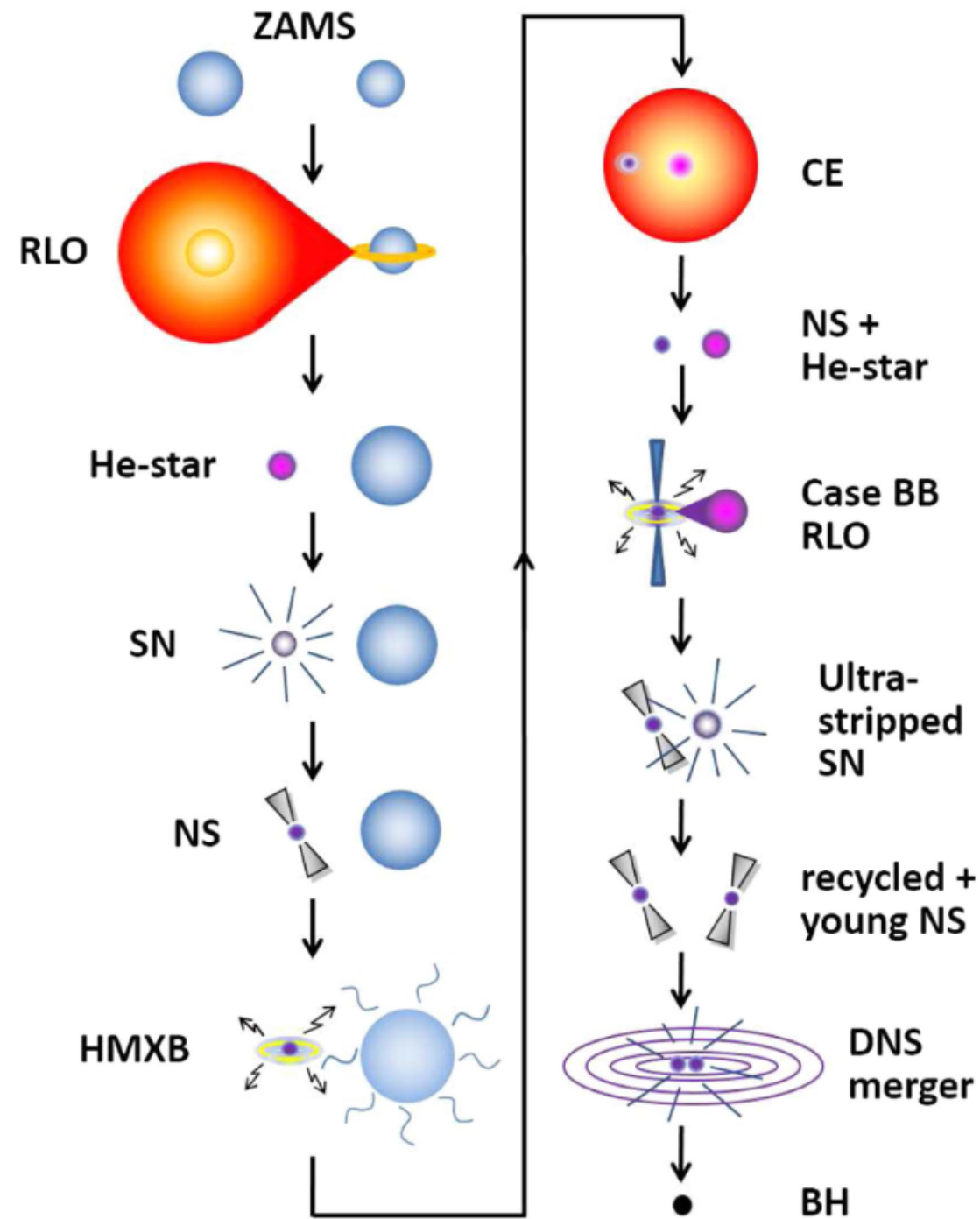
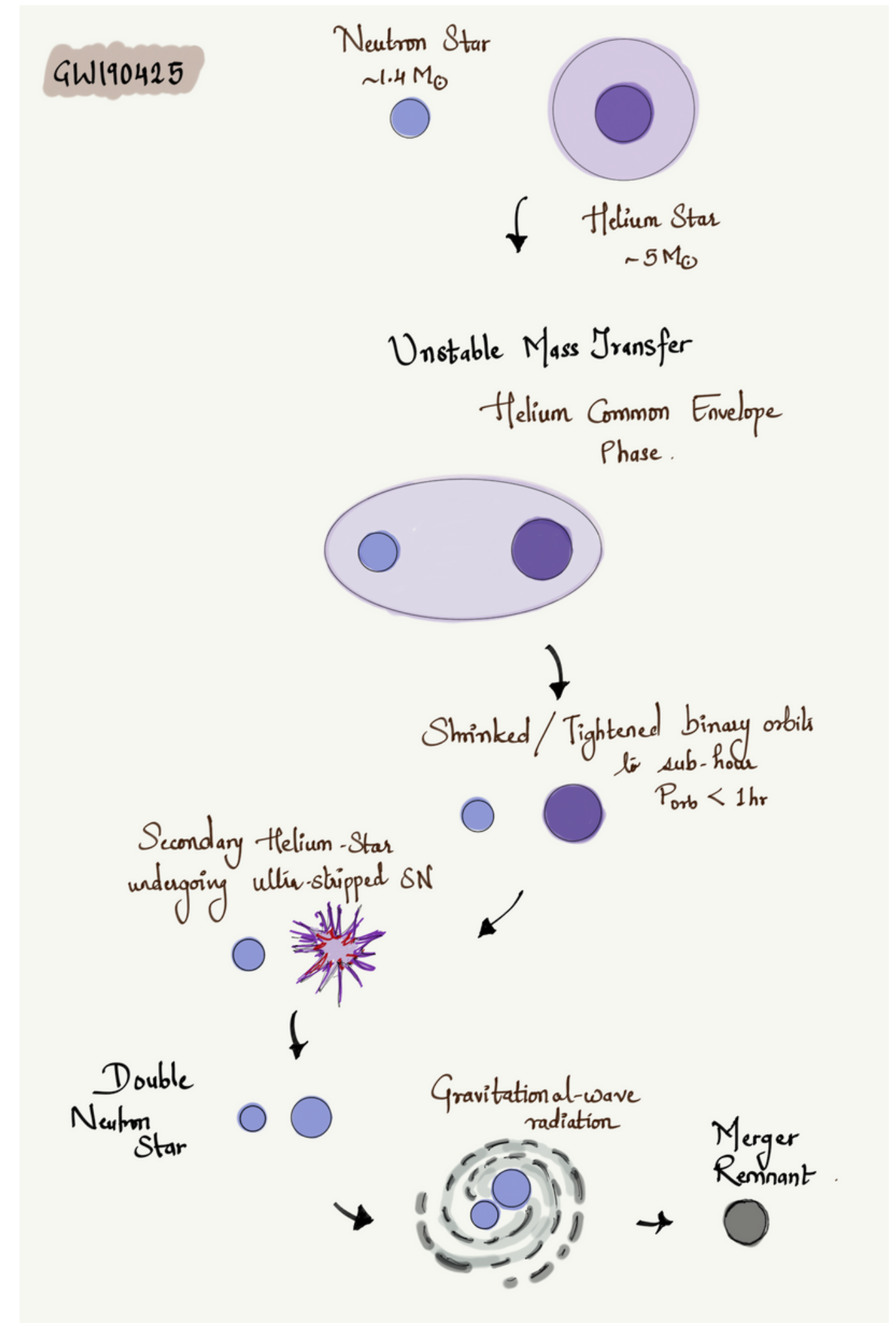


Figure 1. Illustration of the formation of a DNS system that merges within a Hubble time and produces a single BH, following a powerful burst of GWs and a short GRB. Acronyms used in this figure—ZAMS: zero-age main sequence; RLO: Roche-lobe overflow (mass transfer); He-star: helium star; SN: supernova; NS: neutron star; HMXB: high-mass X-ray binary; CE: common envelope; BH: black hole.

- Most systems are expected to survive the second SN explosion.
- If the post-SN orbital period is short enough (eccentricity is large), the DNS system will eventually merge due to GW radiation.
- The final remnant is most likely a BH, although, depending on the EoS, a NS may be left behind instead

Possible formation channel for GW190425

Assuming an Isolated Binary Formation Channel



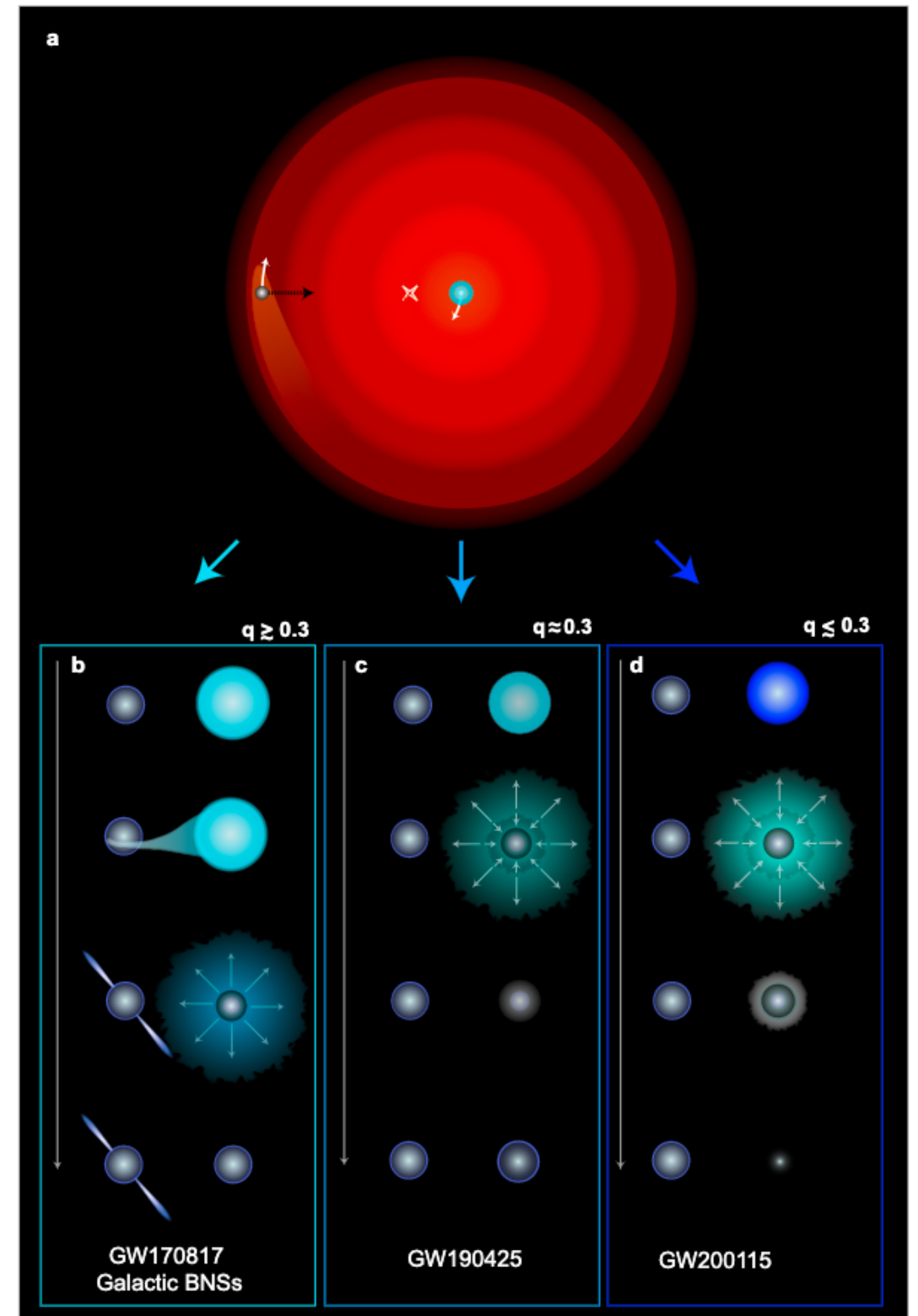
Massive Helium stars

A massive ($\geq 9 M$) helium stars remain compact and avoid mass transfer onto a neutron star and thus avoiding pulsar recycling.

Massive helium stars could lead to a radio-quiet compact binaries that can only be detected by gravitational waves.

This can explain the dichotomy between the observed BNSs hosting recycled pulsars, GW170817 and heavy gravitational wave source GW190425

Vigna-Gomez et al. (2021)



Alternate Proposed Formation Channel (Dynamical Interaction)

- Replace the stellar companion of massive NS by a NS through dynamic encounter.
- Dynamical formation channel proposed for three BNS in the galactic field with short orbital periods and high eccentricities to the globular cluster DNS PSR B2127+11C - formed in globular clusters, but ejected into the field due to dynamic interaction

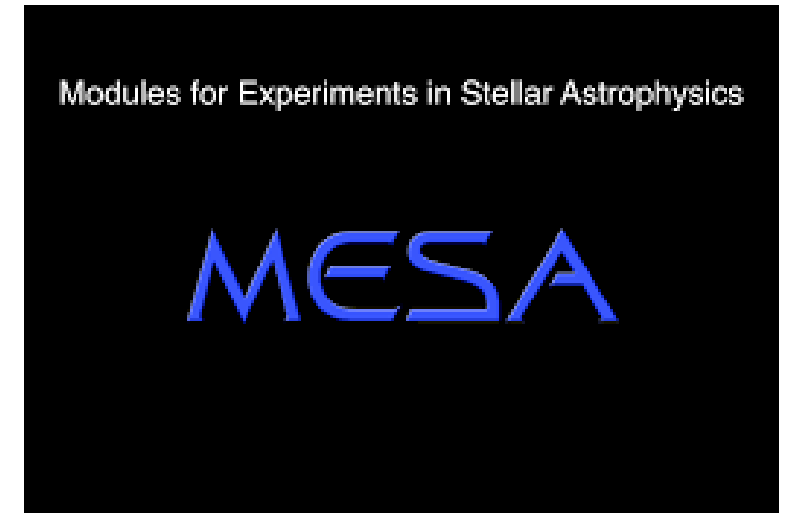
Future Goal

Theoretical predictions for the properties of GW merger sources by using MESA (Modules for Experiments in Stellar Astrophysics) to model the evolution of massive stars (binaries), to investigate the formation channel of massive merger sources like GW190425

Along with COMPAS to study the ever growing population of DNS mergers

We hope to detect more DNS sources with the ongoing O4 run!
fingers crossed

This will shed some light on the astrophysical processes such as common envelope phase and evolution of massive stars



<https://docs.mesastar.org/en/release-r23.05.1/>



compas.science

Thank you!

What are the crucial physical processes that decide the final fate of the DNS binary sources?

- Mass and composition of the stripped helium star
- Orbital separation/ period
- Kick imparted on the new-born NS
- Eccentricity
- Metallicity
- Mass transfer : Stable/Unstable
- Recycled/Non-recycled pulsar
- SNe: Stripped SN/ Ultra stripped SNe
- Mass and composition of the stripped helium star after the envelope is ejected.